J Food Sci Technol
https://doi.org/10.1007/s13197-024-05940-2

ORIGINAL ARTICLE

Red dragon fruit-soy protein isolate biofilm: UV-blocking, antioxidant & improved mechanical properties for sustainable food packaging

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Revised: 13 January 2024 / Accepted: 29 January 2024
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Abstract In this study, an active biofilm was developed by incorporating red dragon fruit peel (RDF) extract into soy protein isolate (SPI) film matrix for sustainable food packaging. The addition of betalain-rich-RDF extract (1–7 wt%) significantly improved UV-blocking and antioxidant properties of the film compared to the control film. As wt% of RDF-extract increased, water vapor permeability, water solubility, and elongation at break decreased by 1.06×10^{-10} g m m^{-2} s^{-1} Pa^{-1}, 34.25%, and 133.25%, respectively. On the other hand, Tensile strength increased significantly ($P < 0.05$) by 78.76%. FTIR results confirmed the intermolecular interaction between RDF extract and SPI through hydrogen bonding, while XRD result showed a decrease in the crystallinity degree of the film with RDF extract addition. However, no significant change in the TGA curve between extract-incorporated SPI films was observed. SEM analysis revealed that SPI B and SPI D films had a more compact and denser structure than the control film, while AFM analysis showed an increase in R_a and R_q values representing higher surface roughness of SPI D film. SPI D film also significantly ($P < 0.05$) decreased the weight loss and increased total soluble solids of freshly cut apples over 7-day storage period.

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/s13197-024-05940-2.

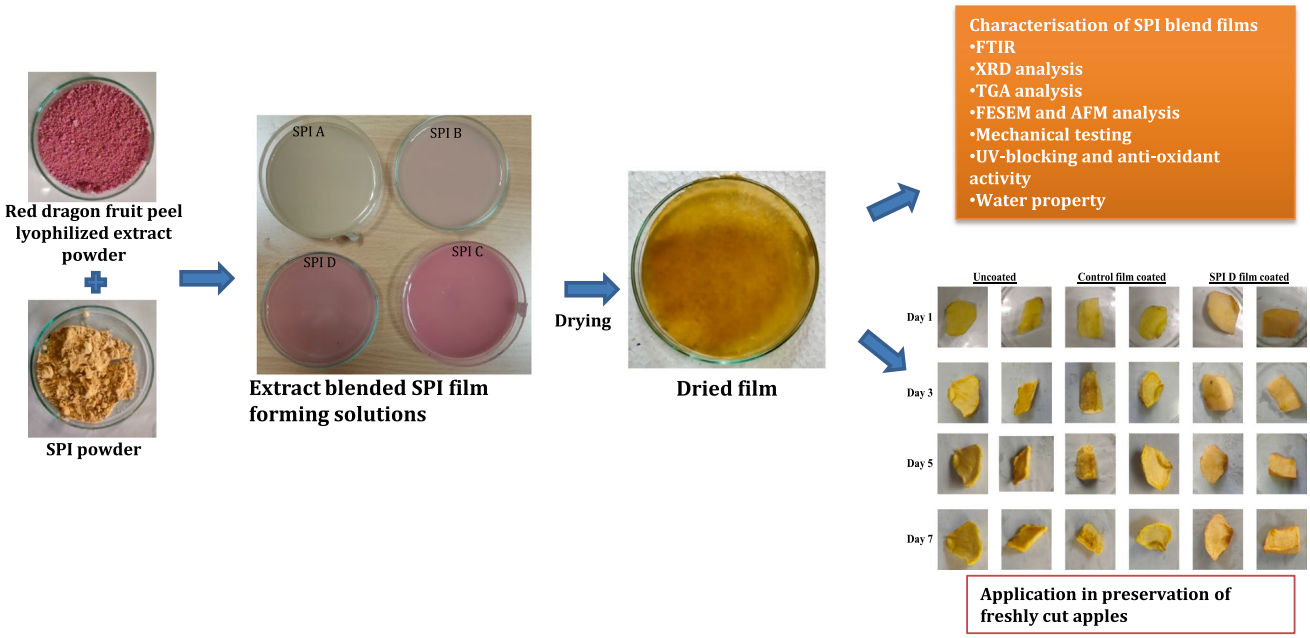
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Published online: 03 March 2024

Graphical abstract



Keywords Bio-active film · Soy protein isolate · Red dragon fruit peel waste · Edible coating · Antioxidant film

Introduction

Using petroleum-based plastics has caused several problems like dependence on petrochemical feedstock, environmental pollution, and pressure on landfills. Therefore, there is a need to develop novel biodegradable packaging material made from natural polymers to ease the effect of pollution (Koshy et al. 2015). Due to the fundamental property of degradation in natural climate, biodegradable polymers have received considerable attention from researchers.

Soy protein isolate (SPI) is often employed to fabricate films and biocomposites that can be used for edible purposes. SPI is manufactured as a byproduct of defatted soybean flour during the production of soybean oil (Zhou et al. 2021). SPI is cheaper and widely available as well. Two types of globulin proteins are found in SPI: (a) beta-conglycinin and (b) glycinin, with a molecular weight of 150 kDa and 350–400 kDa (Zhou et al. 2021).

Multiple varieties of cross-linking agents are added to SPI films to enhance their performance. Limited literature was found in which various extracts from fruits and vegetable waste are used as cross-linking agents like phenolic compounds (Friesen et al. 2015), grape seed extract (Sivarooban et al. 2008), licorice residue extract (Han et al. 2018), and mangosteen peel extract (Zhou et al. 2021). The purpose of adding extract into the SPI film was to increase antimicrobial activity, antioxidant activity, mechanical properties, UV-blocking properties, etc. (Friesen et al. 2015).

Dragon fruit is an exotic tropical fruit with a sweet taste. The peels contribute to 30–35% of total fruit weight. They are generally discarded as waste which creates environmental problems. However, dragon fruit peel contains phenolic compounds, betalain compounds, pectin, etc., which are known to have nutritional value. Betalains in red dragon fruit peel are suitable food colorants and have excellent antioxidant, anti-cancer, anti-microbial, and anti-lipidemic activity (Bhagya Raj and Dash 2020). To the best of our knowledge, no literature has been found in which red dragon fruit peel extract was incorporated into SPI film.

The thin layer of edible coating is created over a food product by spraying or dipping method. The objective of edible coatings and films is to extend the storage life of food products. In literature, SPI-based film and coating have been used to improve the shelf life of fresh-cut eggplant, whole apricot, freshly cut apple, blueberry (Alves et al. 2017) etc. It was observed that extract-incorporated SPI film is better at controlling weight loss, color, and firmness of freshly cut apples than uncoated apples (Alves et al. 2017).

Migration tests are employed to indicate compliance with regulatory legislation and ensure customer health. The overall migration test checks how inert a substance is. Specific migration test is for purposefully added additives. According to commission regulation (EU) No. 10/2011, plastic packaging that comes into contact with food should have an overall migration limit of 60 mg/kg of food. The specific migration limit (SML) for each specific substance is well established based on its toxicity. In the commission regulation (EU) No. 10/2011, the specific migration limit of total phenolic content is not mentioned. Therefore, for the substance for which the no SML, a general limit of 60 mg/kg of food is considered (Agustinelli et al. 2021).

For the browning process of the apples, the polyphenol oxidase (PPO) enzyme is mainly responsible. When the enzyme and substrate come into contact with each other, the browning process starts. Oxidation of phenolic compounds is catalyzed by the PPO, resulting into the formation of melanin, and producing the brown pigment on the surface of freshly cut apples. After peeling and cutting the apples, an increase in PPO activity is expected. Previous studies involving the preservation of pears have shown the correlation between PPO activity, CO_2 concentration in package headspace atmosphere, and CIE L, a, and b values measured using a colorimeter (Soliva-Fortuny et al. 2002). Preservation of fresh-cut apples using whey protein isolate film incorporated with citric acid and montmorillonite was carried out. This study also found that the browning index with PPO activity and color values (L, a, and b values) correlate (Azevedo et al. 2018). PPO activity was directly correlated with the browning index for minimally processed apples (Rocha and Morais 2002). In the current study, apples were classified into three groups: (1) uncoated apples (2) SPI film coated apples, and (3) SPI D film coated apples. Then, the PPO activity of freshly cut apples was measured at regular intervals of 1, 3, 5, and 7 day.

The novelty of the current work lies in the utilization of the red dragon fruit peels as a waste product for the extraction of valuable compounds and their incorporation into the soy protein isolate based biofilms. By addressing two major environmental problems, namely fruit peel waste disposal and synthetic plastic packaging, the current work offers a comprehensive solution with significant implications for the food industry. Red dragon fruit peels are about 30–35% of the fruit's weight. Their improper disposal leads to the loss of valuable compounds and environmental challenges. The highly concentrated red dragon fruit peel extract powder was obtained by extraction followed by freeze-drying. Our findings demonstrate that extract incorporation enhances several properties of the biofilm. Moreover, the application of RDF extract incorporated SPI film for the shelf-life enhancement of freshly cut apples is reported. This approach addresses plastic packaging concerns and provides a cost-effective solution. By utilizing the waste from the fruit industry and adopting the cheaper and widely available soy protein isolate, the current study is an economically viable alternative to synthetic plastic packaging. This cost-effective solution has the potential to broadly impact the food packaging industry, mitigating environmental problems associated with traditional plastic packaging.

The hypothesis of the current research was to improve the property of the SPI-based edible film by incorporating red dragon fruit peel (RDF) extract. The fabrication and characterization of modified SPI film were carried out using different techniques. Furthermore, the application of RDF peel extract incorporated SPI film in preserving and improving fresh-cut apple shelf life was carried out. For this purpose, weight loss, total soluble solids, and physical appearance were chosen as essential parameters to measure the quality of freshly cut over 7 days at room temperature. Uncoated freshly cut apples under analysis were chosen as the control.

Materials and methods

Raw materials and chemicals

Red dragon fruits, apples, and soy protein isolate (90% protein) were purchased from a local market in Surat, India. Glycerol (98%) was acquired from Finar limited in Ahmedabad. Standard betanin, HPLC grade acetonitrile, trifluoroacetic acid, and water were procured from Sigma-Aldrich Chemical Co. Ltd.

Extraction of dragon fruit peel

The freeze-dried red dragon fruit peel powder was prepared using previously described method (Prajapati and Jadeja 2023). The peel powder was subjected to extraction using water as solvent, 10 g of lyophilized red dragon fruit peel powder was mixed with 200 mL of water, stirring overnight, filtered using Whatman filter paper no.1, and the supernatant was subjected to centrifuge at 9000 rpm for 10 min. The supernatant was then concentrated using a rotary evaporator, frozen overnight, and finally freeze-dried for 6 h. The lyophilized extract was ground into powder and stored in polyethylene ziplock bag at − 4 °C.

Total phenolic, total betacyanin, and total betaxanthin content of RDF extract

Total betacyanin and betaxanthin content were determined using a spectrophotometric method as described in the literature (Sharma et al. 2021). A known quantity of extract

was diluted in deionized water before analysis, and the following equation were used to express the results in mg per g of lyophilized extract.

$$Total\ betacyanin\ content = \frac{A_{536} \times DF \times MW_1 \times 1000}{60000 \times i} \quad (1)$$

$$Total\ betaxanthin\ content = \frac{A_{480} \times DF \times MW_2 \times 1000}{48000 \times i} \quad (2)$$

where, DF is dilution factor, i is path length in cm, MW_1 and MW_2 are the molecular weight of betacyanin (550 g/mol) and betaxanthin (339 g/mol), A_{536} and A_{480} represent the absorbance at 536 nm and 480 nm, molar extinction co-efficient (ε) for betacyanin and betaxanthin are 60,000 mol^{-1} cm^{-1} and 48,000 mol^{-1} cm^{-1}, respectively.

Total phenolics content (TPC) was measured using a modified Folin-Ciocalteu (FC) method (Mahayothee et al. 2019). One milliliter of appropriately diluted sample was mixed with 1 mL of 0.1 M FC reagent, and incubated for 3 min, and then mixed with 1 mL of 0.5% Na_2CO_3 solution for 30 min in the dark. Using a UV–visible spectrophotometer (SPECTROstar nano, BMG labtech Ltd., Germany), the absorbance was measured at 765 nm, and the values of TPC were expressed as Gallic acid equivalent using the following equation:

$$TPC = \frac{C \times V \times D}{M} \quad (3)$$

where, C is the concentration of TPC in ppm, V is the solvent volume, D is the dilution factor, and M is the weight of the material.

HPLC analysis of the extract

Betanin compounds present in the lyophilized RDF extract were quantified using HPLC analysis. 0.1 g of RDF extract was dissolved in 20 mL deionized water for the analysis. HPLC method employed is descibed in prevous literature (Prajapati and Jadeja 2023).

Fabrication of SPI films

SPI films were fabricated using the method described elsewhere (Koshy et al. 2022). 2.5 g of SPI was stirred in 50 mL water. Then, 1 mL glycerol (40% w/v) was added as a plasticizer. The appropriate quantity of 2 N NaOH solution was added to maintain neutral pH. The mixture was then heated to 80 ℃ for 30 min to support the denaturation of protein. The appropriate amount of extract was added, and the solution was then sonicated to remove the bubbles. Finally, 18 mL of solution was cast into a 10 cm^2 diameter petri dish and kept at room temperature overnight. The composition of films is tabulated in Table S1. As shown in Table S1, different films are formulated like, control film (film without extract), SPI A film (1% extract), SPI B film (3% extract), SPI C film (5% extract), and SPI D film (7% extract). Fig. S1 shows the films fabricated using the above technique.

Antioxidant activity

DPPH scavenging assay was carried out to measure antioxidant activity (Qin et al. 2020a). Briefly, various amounts of film samples (0, 4, 12, 20, 28, and 36 mg) were combined with 4 mL of methanolic DPPH (0.06 mM) solution and left at room temperature for 1 h in the dark. Absorbance was measured against a blank solution at 517 nm.

Optical properties of the biofilm

A microplate reader (SPECTROstar nano, BMG labtech) was used to measure the transmittance spectra of films by scanning the wavelength between 200–800 nm. Before testing, biofilms were cut into 10 × 40 mm pieces and placed directly into the test cell of a spectrophotometer. An empty cell was taken as the blank reading.

Mechanical testing

Using Instron-5967 universal testing equipment, the tensile strength (TS) and % elongation at break for a film specimen (120 × 10 mm) were measured. The test was performed at crosshead speed of 0.8 $mm{\cdot}s^{-1}$. The initial gauge length was 60 mm. The portable dial thickness gauge was used to measure thickness with a precision of 0.01 mm. The mean of thickness was obtained from four random measurements.

Moisture content (MC)

The film samples were dried in an air oven at 110 ℃ for 24 h (Xiao et al. 2020). MC was quantified using the following equation:

$$MC(\%) = \frac{M_1 - M_2}{M_1} \times 100 \quad (4)$$

where, M_1 and M_2 are the initial and final weights, respectively.

Water solubility (WS)

WS of SPI films was measured using a method that has been previously reported (Xiao et al. 2020). The 30 × 30 mm films

were initially weighed (M_i) and immersed in 40 mL water for 24 h. After that, film was dried overnight at 50 ℃, and the final weight was measured (M_f). WS was calculated using the following formula:

$$WS(\%) = \frac{M_i - M_f}{M_i} \times 100 \quad (5)$$

where, M_i and M_f are the initial and final weights, respectively.

Water vapor permeability (WVP)

WVP was measured using the gravimetric method (Zhang et al. 2019). 10 g of anhydrous $CaCl_2$ was placed in a glass cup of 1.2 cm^2 diameter, and biofilm was sealed at the top. Then, it was placed in a desiccator at room temperature and 90% relative humidity. An increase in the weight of $CaCl_2$ was measured after 24 h. WVP was quantified using the following formula:

$$WVP = \frac{\Delta m \times z}{A \times t \times \Delta P} \quad (6)$$

where, Δm is the increase in weight of $CaCl_2$ (g), z is the thickness (m), A is the effective area of the film (m^2), ΔP is the partial pressure of the water (Pa) and t is time (s).

FTIR

FTIR measurement were carried out by a Shimadzu (IRAffinity-1 s) spectrometer. Each spectrum was collected in wavelength range of 500–4000 cm^{-1} as average of 45 scan and 4 cm^{-1} resolutions. A numerical deconvolution of the Amide-I region was carried within the spectra range of 1600–1700 cm^{-1}. A linear baseline was generated for each region. Also, the absorbance was normalized in relation to the maximum peak so that unwanted intensity changes can be avoided. The initial values for the peak position were found using Fourier deconvolution. All the calculations to obtain the relative area for deconvolution were performed in MS Excel.

FESEM

FESEM (Nova Nano, FESEM 450, FEI) was carried out to observe the surface morphology. Au coating was carried out with a cycle time of 10 s and a coating thickness of 2–5 nm. For cross-sectional FESEM images, the samples were immersed into liquid nitrogen and cut according to the requirement.

AFM

The surface roughness of the SPI films were measured using multitube 8 AFM (Bruker, Germany). The analysis was done by capturing the AFM images and then evaluating the average roughness (R_a) and root mean square roughness (R_q) using the Gwyddion software.

XRD

The crystal phase and crystalline structure of the SPI films were measured using XRD analysis (Rigaku Miniflex, Cu Kα radiation 1.54065). XRD patterns were recorded from 5° to 50° (2θ) at a scan rate of 5°/min.

TGA

The thermal properties of the SPI films were analyzed by a thermal gravimetric analyzer (NETZSCH TG 209F1 libra). Approximately 50 mg sample was heated in a temperature range from 30 to 800 ℃ with a ramp up rate of 20 ℃/min under a nitrogen environment.

Overall migration test (OMT)

Since the application of SPI film coating in apple preservation is performed at 4 ℃, the OMT and Specific migration test (SMT) were both performed at 4 ℃ and room temperature. Double sided total immersion method provided in EU 10/2011, OM tests were carried out to evaluate the release of all the substances that can migrate into two liquid food simulants: 5% ethanol (simulant A for aqueous food) and isooctane (Simulant D2 for fatty food). 10 dm^2/L Film strips were accurately weighed in an analytical balance (Mettler Toledo, accuracy ± 0.0001 g) and placed in a glass vial containing food simulant fluid, and kept at 4 ℃ and room temperature. The films were kept in isooctane simulant for 10 days (Agustinelli et al. 2021). To maintain dimensional integrity and to avoid any physical changes during the study, the contact time for films immersed in 5% ethanol was kept at 3 days (Kuorwel et al. 2013). The overall migration was determined by evaporating the simulants, and migrated compounds were quantified gravimetrically.

Specific migration test (SMT)

6 dm^2/L Films strips were immersed in glass vials containing 5% ethanol. The test was performed at 4 ℃ and room temperature. The contact time for 5% ethanol was 3 days (Agustinelli et al. 2021). The films were then removed,

and the simulant fluid was filtered using a syringe filter (0.45 µm). Previous literature reported the TPC analysis of simulant fluids during migration study. Based on the method reported, the TPC analysis was carried out (Chong et al. 2021).

Application in the preservation of freshly cut apples

Freshly cut apples were used to test SPI film preservation properties. Apples of similar size and maturity were purchased from the local market. After peeling, they were cut into similar size pieces ($3 \times 4 \times 2$ cm^3). Three coating tests were conducted: group 1 (uncoated apples), group 2 (coating with control film solution), and group 3 (coating with SPI D film solution). In each group, 6 freshly cut apples were present. Each freshly sliced apple was submerged in a film-forming solution for one minute and then air-dried at room temperature (Alves et al. 2017). All samples were stored at 4 ℃ for 7 days.

Weight loss (%) and physical appearance of apples

The weight loss was calculated gravimetrically and calculated as a percentage of the initial weight (Liu et al. 2017). Each apple was photographed and visually observed for 7 days as well.

Soluble solids content (SSC) of apples

The content of total soluble solids of the freshly cut apples was measured using Abbe refractometer (Model-2501-G), and the results were expressed in % (Alves et al. 2017). SSC was measured on day 1 and compared with data on day 7.

Measurement of PPO activity

The enzyme extraction of apple slices was carried out using the method developed in previous literature (Pilon et al. 2015). 20 mL of 0.2 M sodium phosphate buffer (pH 7) and 10 g of apple slices were grounded in a high speed mixer (Blender 7010S; Waring, Torrington, India). The resulting homogenised solution was centrifuged at 12,000 rpm for 30 min. The supernatant was then filtered through Whatman filter paper No.1. The PPO activity was determined by mixing 2.9 mL of 0.11 M catechol in 0.05 M phosphate buffer (pH 7) and 0.1 mL enzyme extract and measuring the increase in absorbance at 420 nm in a microplate reader (SPECTROstar nano, BMG labtech). The measurement was carried out for 5 min time and absorbance was measured

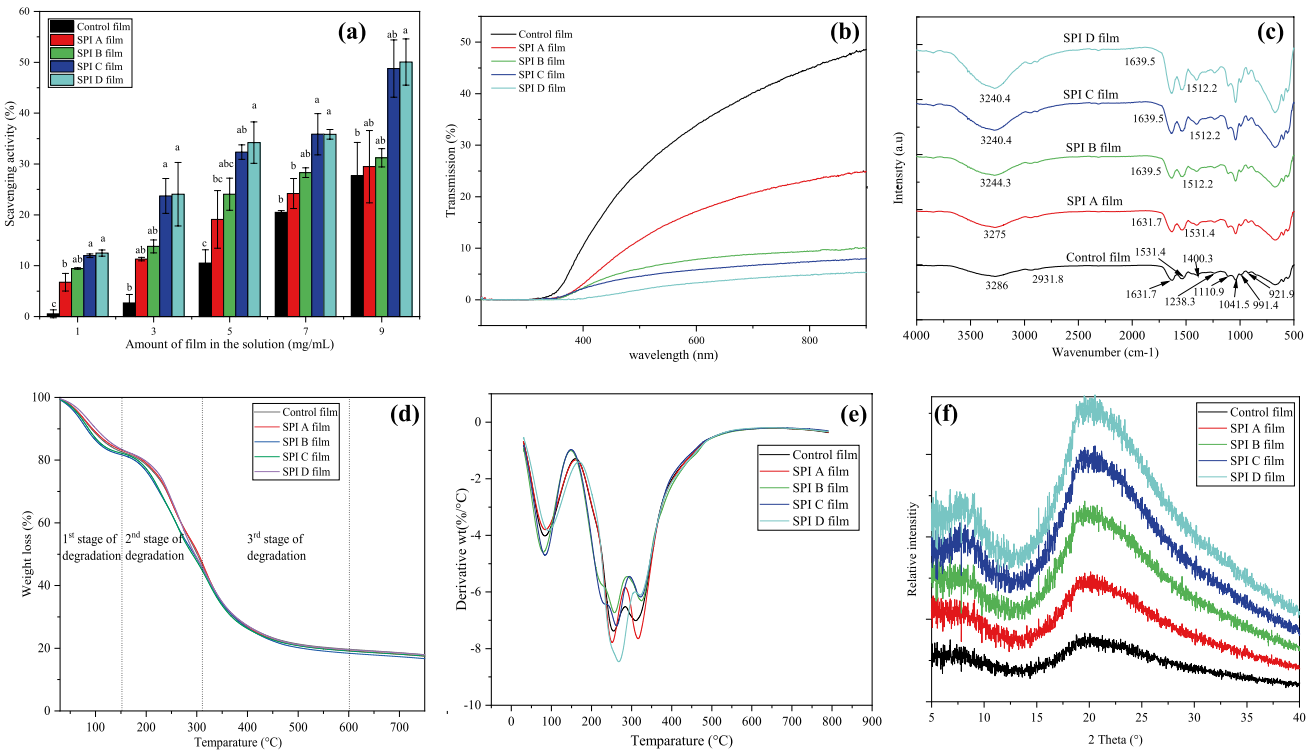


Fig. 1 **a** DPPH radical scavenging activity, **b** UV–vis spectra, **c** FTIR, **d** TGA, and **e** DTG and **f** XRD of SPI, SPI A, SPI B, SPI C, and SPI D film. For Fig. (**a**), the values provided are the average ± standard deviation, calculated from a sample size of 2. According to tukey test, different letter in the same column represents significant difference with a *p*-value less than 0.05

every 15 s. The linear portion of absorbance vs time curve was taken into account. One unit of PPO activity is defined as increased absorbance in 0.001 min^{-1} ml^{-1} after instantly adding the enzyme extract.

Statistical analysis

The statistical analysis was carried out using Minitab 18. The data were analysed using the one-way analysis of variance (ANOVA) followed by Tukey's test. In all the tables, different letter in the same column represents significant difference with a p-value less than 0.05. The reported values represent the average ± standard deviation, which were calculated from a sample size of 3.

Result and discussion

Characterization of RDF extract

The extraction yield was found to be 0.1 g/g of lyophilized powder. TPC of the lyophilized extract was found to be 67.93 mg Gallic acid equivalent/g of extract. Betanin and isobetanin found using HPLC analysis were 208.66 and 13.6 mg/g of extract, respectively. On the other hand, total betacyanin and total betaxanthin content found using the spectrophotometric method were 245.51 and 189.15 mg/ g of extract, respectively. In the literature, the total betacyanin content of RDF extract vacuum-dried powder was reported to be 175 mg/g extract (Qin et al. 2020b). The main betacyanin found in RDF extract is betanin, isobetanin, phyllocactin, isophyllocactin, butyrybetanin, and iso-butyrybetanin (Qin et al. 2020a). Furthermore, the main betaxanthin present in RDF extract is indicaxanthin and isoindicaxanthin (Huang et al. 2021). The reason for the difference in our value of total betacyanin content and literature value could be because of discrepancies in the growing environment, different varieties, extraction methods, and solvent utilized for red dragon fruit (Li et al. 2022).

Antioxidant activity of biofilm

As shown in Fig. 1a, the addition of RDF significantly ($P < 0.05$) increased the film capacity to scavenge DPPH radicals. The highest antioxidant activity was found for SPI D film, with a significant increase as compared to the control film. The reason for this can be attributed to the betacyanin compound and phenolic compounds present in the peel extract, which has antioxidant activity (Maryam Adilah and Nur Hanani 2019; Qin et al. 2020a). A similar trend of increase in anti-oxidant activity was found in literature, in which different betacyanin-rich powder or extract was used for the fabrication of different films (Aparicio-Fernández et al. 2018; Maryam Adilah and Nur Hanani 2019). This result represents the excellent potential of RDF extract-SPI blend film as an antioxidant packaging material.

Optical property of biofilm

UV-blocking is one of the essential properties of the packaging material, as exposure to UV light may lead to quality problems in photosensitive products like oxidization, nutrient loss, and flavor loss. UV–vis spectra of all SPI films are shown in Fig. 1b. All films exhibited almost zero transmittance in 200–300 nm range. This could be because of the presence of aromatic amino acids in the protein (Zheng et al. 2017). All the films incorporated with extract showed low transmittance in the 300–400 nm range and visible range of 400–800 nm. This could be attributed to the presence of betacyanin pigment and phenolic compounds in the extract. Improvement in UV-blocking behavior of biofilms was observed in literature as well (Han et al. 2018).

WVP of biofilms

WVP represents the water vapor barrier property of the SPI films, and a low WVP value is desirable for application in food packaging. The WVP of biofilms in the absence and presence of RDF extract can be observed from Table 1. The

Table 1 Thickness, tensile strength (TS), % elongation at break, moisture content (MC), water solubility (WS), water vapor permeability (WVP) of control film, SPI A film, SPI B film, SPI C film and SPI D film

	Thickness (mm)	TS (MPa)	% elongation at break	MC (%)	WS (%)	WVP × 10^{-10} (g m m^{-2} s^{-1} pa^{-1})
Control film	0.0192 ± 0.0004^{a}	1.46 ± 0.09^{b}	185.89 ± 2.09^{a}	41.44 ± 2.08^{a}	42.68 ± 3.12^{a}	1.23 ± 0.07^{a}
SPI A film	0.0195 ± 0.0025^{a}	1.19 ± 0.16^{b}	150.80 ± 5.29^{b}	38.01 ± 2.38^{a}	38 ± 3.09^{ab}	1.17 ± 0.06^{ab}
SPI B film	0.021 ± 0.0021^{a}	2.42 ± 0.37^{a}	131.27 ± 4.64^{c}	36.46 ± 2.05^{a}	35.32 ± 2.57^{b}	1.19 ± 0.06^{ab}
SPI C film	0.022 ± 0.0055^{a}	2.45 ± 0.13^{a}	133.01 ± 7.96^{c}	30.16 ± 2.11^{b}	35.14 ± 2.89^{b}	1.07 ± 0.05^{b}
SPI D film	0.0223 ± 0.0033^{a}	2.61 ± 0.36^{a}	133.25 ± 3.93^{c}	28.71 ± 2.8^{b}	34.25 ± 2.47^{b}	1.06 ± 0.06^{b}

control film had the highest WVP value ($P < 0.05$), which was 1.23×10^{-10} g m m^{-2} s^{-1} Pa^{-1} and was comparable to that (1.96×10^{-10} g m m^{-2} s^{-1} Pa^{-1}) obtained in the literature (Wu et al. 2021). The WVPs of SPI A film, SPI B film, SPI C film, and SPI D film were decreased by 4.87%, 3.25%, 13.00%, and 13.82%, respectively. With an increase in addition of RDF extract, the value of WVP was found to be decreasing ($P < 0.05$) to 1.06×10^{-10} g m m^{-2} s^{-1} Pa^{-1}. This may be because of the decrease in water affinity of the film as the polyphenols, and betacyanin compounds penetrated the inner space of the SPI films and thereby creating a restrictive path for water molecules (Yao et al. 2021). Furthermore, RDF extract may have formed a hydrogen bond with SPI and the free water trapped inside the film matrix. These interactions may be responsible for the decreasing inter-chain space between the polymers. Also, the movement of water molecules is decreased. This could decrease the water migration in the film matrix (Friesen et al. 2015). It is evident from the increase in the value of TS that a stronger network structure is formed. Moreover, morphological structural changes are also responsible for changes in WVP apart from chemical structure. The addition of RDF extracts increases matrix compactness, as evident from FESEM results. The result agrees with Qin et al. (2020a, b), who observed a decrease in WVP when betacyanin rich red pitaya peel extract was added to starch/polyvinyl alcohol film. The WVP may depend upon the film matrix type, betacyanin composition, and film compactness.

Mechanical testing

As shown in Table 1, with increasing wt.% of RDF extract, the thickness of SPI blend films also increased. This is mainly due to the incorporation of polyphenolic compounds and betalain compounds. A similar improvement in thickness was obtained in grape seed extract (Sivarooban et al. 2008) and mangosteen peel extract (Zhou et al. 2021) incorporation into SPI. An increase in thickness can be mainly attributed to cross-linking of phenolic and betalain compounds with protein in SPI by hydrogen bonding. This prevents the ordered structure, which increases thickness (Maryam Adilah and Nur Hanani 2019).

TS of the control film was 1.46 ± 0.09 MPa, which significantly increased ($P < 0.05$) by 78.76% to 2.61 ± 0.36 MPa for SPI D film. A similar trend was observed in the literature as well (Kang et al. 2016; Han et al. 2018). The phenolic compounds stabilize the SPI film structure by contributing active groups for intermolecular cross-linking. This increases the rigidity of the film. Therefore, elasticity reduces and TS increases. The improvement in TS is because of this intermolecular cross-linking (Sivarooban et al. 2008). Moreover, the small amount of extract results in low hydrogen bonding with protein. This inhibits the interaction of molecules with protein (Zhou et al. 2021). Therefore, a reduction in TS (1.19 ± 0.16 MPa) and improvement ($P < 0.05$) in % elongation at break ($150.80 \pm 5.29\%$) was observed for SPI A film.

Moreover, it was observed that the value of MC decreased. For the control film, MC was found to be $41.44 \pm 2.08\%$. With increasing the RDF extract, the MC decreased significantly ($P < 0.05$) upto $34.25 \pm 2.47\%$. The decrease in MC is because of the modified network structure of the film resulting from the strong interaction between phenolic compounds and SPI (Koshy et al. 2022). A similar trend of decreasing MC was observed in the literature, in which pine needle extract and cellulose nanocrystal was incorporated in SPI film (Yu et al. 2018). MC of the freeze dried dragon fruit peel powder, fresh dragon fruit peel and SPI was 0.63%, 90% and 5.8%, respectively.

Water solubility (WS)

Up to 3% (w/w of SPI) RDF extract, the value of WS decreased. This could be attributed to cross-linking between RDF extract and protein structure, making a three-dimensional intermolecular network. The addition of more RDF extract does not significantly change the value of WS, because protein structure is now attached to one another by means of intermolecular cross-linking, and the quantity of soluble matter remains unaffected. Therefore, on further addition of RDF extract up to 7% (w/w of SPI), WS remained almost unchanged. Similar behavior was observed for values of % elongation at break. % elongation at break decreased up to 3% (w/w of SPI) RDF extract and remained almost constant on further addition of RDF extract. Increasing the extract concentration may restrict the movement between polymer chains, causing more rigidity in the film matrix (González et al. 2011).

FTIR analysis

FTIR analysis was performed to monitor changes in the protein's secondary structure and behavior during interaction with other compound. The control film and RDF-incorporated SPI film showed similar behavior in FTIR spectra as shown in Fig. 1c. The spectra of the SPI film showed peaks at 3286 cm^{-1} (amine N–H stretching and free –OH groups), 1041.56 cm^{-1} (C–H and C–O–H bending), 800–1150 cm^{-1} (glycerol related absorption bands), 2931.8 cm^{-1} (–CH_2 group stretching vibration), 3200–3400 cm^{-1} (O–H and N–H bending vibrations) (Zhang et al. 2012; Xu et al. 2015). The addition of extract caused the peak in the range of 3700–3100 cm^{-1} to become broader, indicating the interaction through hydrogen bonding (Hoque et al. 2010). The major amide bands of SPI were found at 1631.7, 1531.48, and 1400 cm^{-1}, corresponding to C=O stretching (amide I), N–H bending (amide II), and C–N stretching (amide

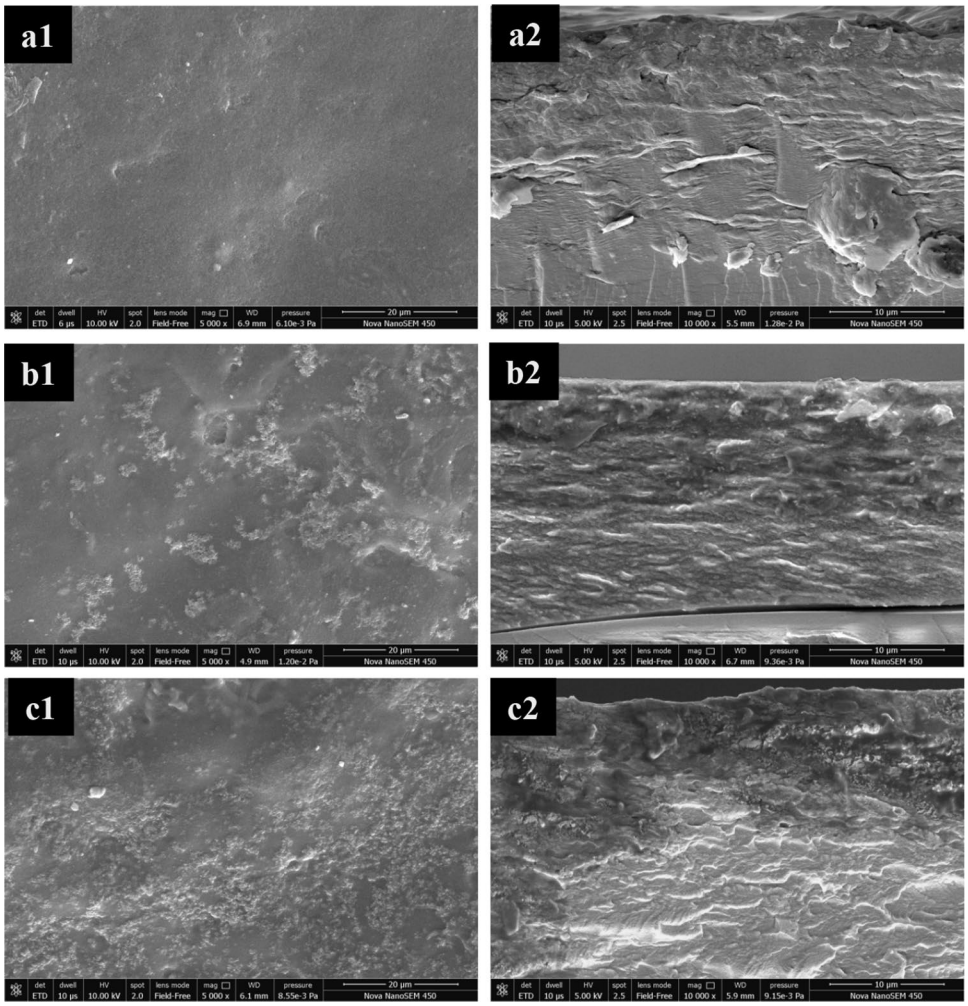


Fig. 2 FESEM images of surface of control film (**a1**), SPI B film (**b1**), SPI D film (**c1**), and cross-section of control film (**a2**), SPI B film (**b2**), SPI D film (**c2**)

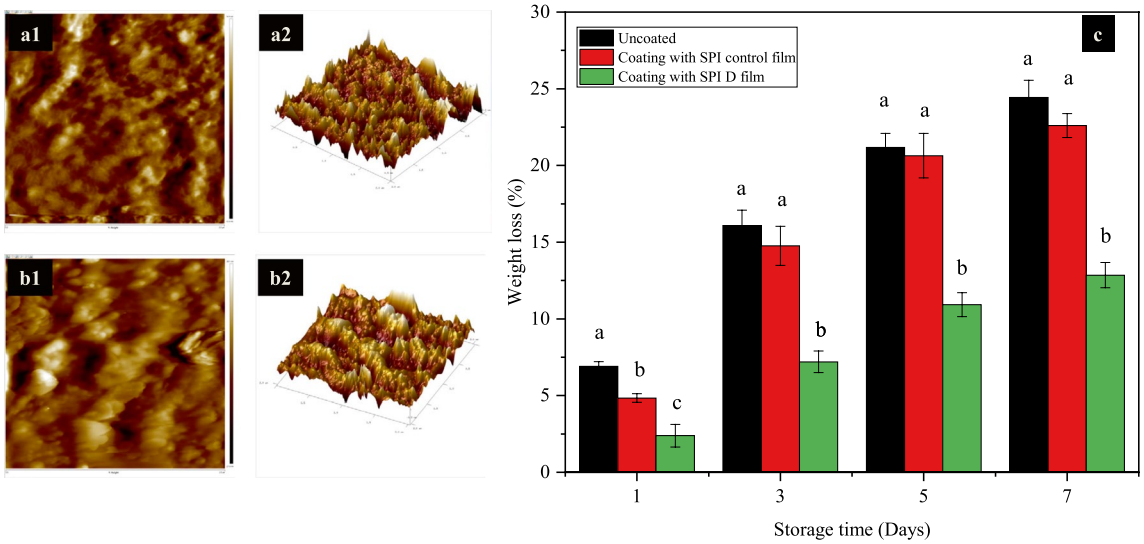


Fig. 3 AFM analysis of control film (**a1**, **a2**), and SPI D film (**b1**, **b2**); Effect of different coating on weight loss (%) of freshly cut apples (**c**), the values provided are the average ± standard deviation, calculated from a sample size of 3. According to tukey test, different letter in the same column represents significant difference with a *p*-value less than 0.05

III), respectively (Li et al. 2016). The peak position shifted from 1631.7 to 1639.5 cm^{-1} for amide-I and 1531.4 to 1512.2 cm^{-1}, potentially indicating interaction with the betalain group and the change in the protein's secondary structure (He et al. 2016). The change in intensity of the amide-I and amide-II bands can be attributed to betalain bonding to N–H, C–N, and C=O groups of protein through hydrophobic interaction (Zhao et al. 2020).

Through deconvolution, four bands were observed within 1600–1700 cm^{-1} spectra region. Fig. S2 represents the deconvoluted spectra of all SPI films. For β-sheet, 1640–1600 cm^{-1} and 1670–1690 cm^{-1} were considered. While for α-helix 1650–1660 cm^{-1} bands and for β-turn, 1700–1660 cm^{-1} bands were taken into account. As shown in Table S2, the percentage area for β-sheet, α-helix and β-turn were found to be 48.72%, 25.44%, and 25.83%, respectively was reported for SPI film. Previous study reported the values of the percentage area for β-sheet, α-helix, and β-turn to be 49.06%, 14.05%, and 27.23%, respectively, for SPI control film (Wang et al. 2016). The film formation technique, % glycerol added during preparation, pH value of the film forming solution, % of extract added, and nature of extract may result in difference in protein unfolding. Therefore, the literature values and trend of the percentage area of β-sheet, α-helix, and β-turn may differ from the current study. With the addition of RDF extract, the increase in conversion of β-sheet was observed. This could be because of RDF extract may have increased the protein unfolding and refolded it into novel β-sheet structure. With the incorporation of 80 g/kg of chestnut extracts into the SPI film, an increase in β-sheet was observed (Wang et al. 2016).

TGA-DTA

The TGA and DTA of the SPI film samples are shown in Fig. 1d, e. In the first stage of degradation, the SPI films showed initial weight loss in the temperature range of 30–150 ℃. This weight loss is mainly attributed to moisture evaporation. The second stage of degradation (150–310 ℃) is mainly attributed to glycerol evaporation and degradation of RDF extract (Kang et al. 2016). As shown in supplementary data (Table S3), the weight loss of SPI A, B, C, and D films was slightly lower than control film in the second stage. This could be because of hydrogen bonding between RDF extract and SPI. Similar weight loss during the second stage of degradation was observed in the literature (Mondal et al. 2022). The third stage of degradation (310–600 ℃) was a result of the thermal disruption of C–C, C–N, and C–O bonds in the soy protein backbone (Li et al. 2016). With increasing concentration of RDF extract into SPI film, no significant change in the TGA curve was observed. It can be concluded from Table S2 that no notable changes in mass loss and T_{onset} occurs between SPI-extract blended films. Zhou et al. (2021) also observed no change in thermal stability when mangosteen peel extracts were incorporated into SPI film. Our result showed that all SPI-blend films studied are thermally stable up to a temperature of 140 ℃, which is favorable for applications in food preservation.

FESEM and AFM

FESEM analysis was used to evaluate the microstructural morphology. Figure 2 represents surface and cross-sectional FESEM images of control film, SPI B film, and SPI D film. All films lacked visible holes and bubbles. Control film showed a relatively smoother surface. On the other hand, the addition of RDF extract led to a coarse surface as well as a more compact and denser structure clearly evident from FESEM images SPI B and SPI D film (Fig. 2). The formation of aggregates is from major cross-linking density zones (González et al. 2011). It might be because of the spherical morphology in SPI B and SPI D is related to cross-linking between RDF extract and SPI. Han et al. (2018) also observed that increasing the addition of licorice residue extract to SPI results in an increase in the roughness of the film.

Figure 3 represents the AFM images of the control and SPI D films, from which the surface morphology was observed. R_q of control film and SPI D film was 19.75 nm and 21.88 nm, respectively, which accounts for a 9.7% increase in R_q. While R_a of SPI and SPI D film was 13.82 nm and 16.28 nm, respectively. Therefore, 14.69% increase in R_a value was observed with increasing RDF extract content in control film. From FESEM images of the surface of biofilms, it is evident that the incorporation of extract has increased the aggregates. This may have contributed to the increase in surface roughness. Xiao et al. (2020) also observed an increase in R_a and R_q values when cellulose nanocrystals and ZnO nanoparticles were reinforced to SPI film. Garrido et al. (2019) also reported an increase in the roughness of SPI blend films reinforced with hydrolyzed keratin.

XRD

XRD patterns of the control film and RDF extract incorporated SPI film were analyzed in Fig. 1f. No sharp peaks were observed for all SPI films, however at low intensity, the control film (SPI film) showed peaks at 2θ values of 6.77° and 19.57°. In the literature, 2θ values of 10.2° and 22° were reported for SPI film (control film); the author reported that these peaks indicated the amorphous nature of the matrix (Koshy et al. 2022). Similar nature was observed for SPI A, SPI B, SPI C, and SPI D films. With increasing concentration of the RDF extract into the SPI film, no new peaks or change in peak position was found. From this, it can be interpreted that there is good compatibility between

RDF extract and SPI. The absence of significant changes in the XRD spectra further supports the notion that the SPI film remains amorphous even after the addition of the RDF extract.

Overall migration test (OM)

SPI is hydrophilic biopolymer, and more susceptible to swelling when it comes in long term contact with aqueous simulants. Therefore, the contact time for OM study should be minimized to reduce the negative impact on the integrity of SPI film. As shown in Table S4, OM for 5% ethanol and isooctane is reported at 4 ℃ and room temperature. OM for isooctane at both temperature conditions was observed to be zero. It was not detected at the quantification level. Isooctane is non-polar in nature and therefore, the SPI film and glycerol may not have interacted with isooctane. According to EU 10/2011, the OM of SPI films for isooctane simulants comply with the regulation limit of 60 mg/kg of simulant. Therefore, there are no limitations when using these films for the application of fatty foodstuffs. Similar observations were made in the literature as well. OM for red grape extract incorporated SPI films was found to be zero (Agustinelli et al. 2021). OM for polylactic acid (PLA)/polyhydroxy butyrate (PHB) film and citrus extract incorporated PLA/PHB film was found to be 6 mg/dm^2 and 7.45 mg/dm^2 (Fiorentini et al. 2021).

For 5% ethanol, OM for SPI films at 4 ℃ was found to be in the range of 1400 ± 80 to 1587 ± 122 mg per kg of simulant. While at room temperature, the OM increased in the range of 1840 ± 280 to 2080 ± 301 mg per kg of simulant. The higher OM values for 5% ethanol as compared to isooctane is mainly because of migrant-matrix affinity and food simulant's ability to dissolve the migrants. With an increase in the temperature, OM values increased as well. This can be attributed to the increase in mobility of protein network, enabling the diffusion of migrants (López-de-Dicastillo et al. 2012). For red grape extract incorporated SPI films, the OM values were found in the range of 1622–2387 mg/kg of simulant for 4–40 ℃ temperature range (Agustinelli et al. 2021).

Specific migration test (SM)

Table S5 shows the SM values of TPC at different temperatures for all SPI films. At 4 ℃ and at room temperature, TPC was found to be in the range of 0.3 ± 0.02 to 1.85 ± 0.66 mg GAE/g of film, and 0.5 ± 0.16 to 1.86 ± 0.11 mg GAE/g of film, respectively. The attractive forces between the polymers or migrants and polymer chains are overcome by the thermal energy provided by the increase in temperature (Paramita and Kasapis 2019). The TPC values were found

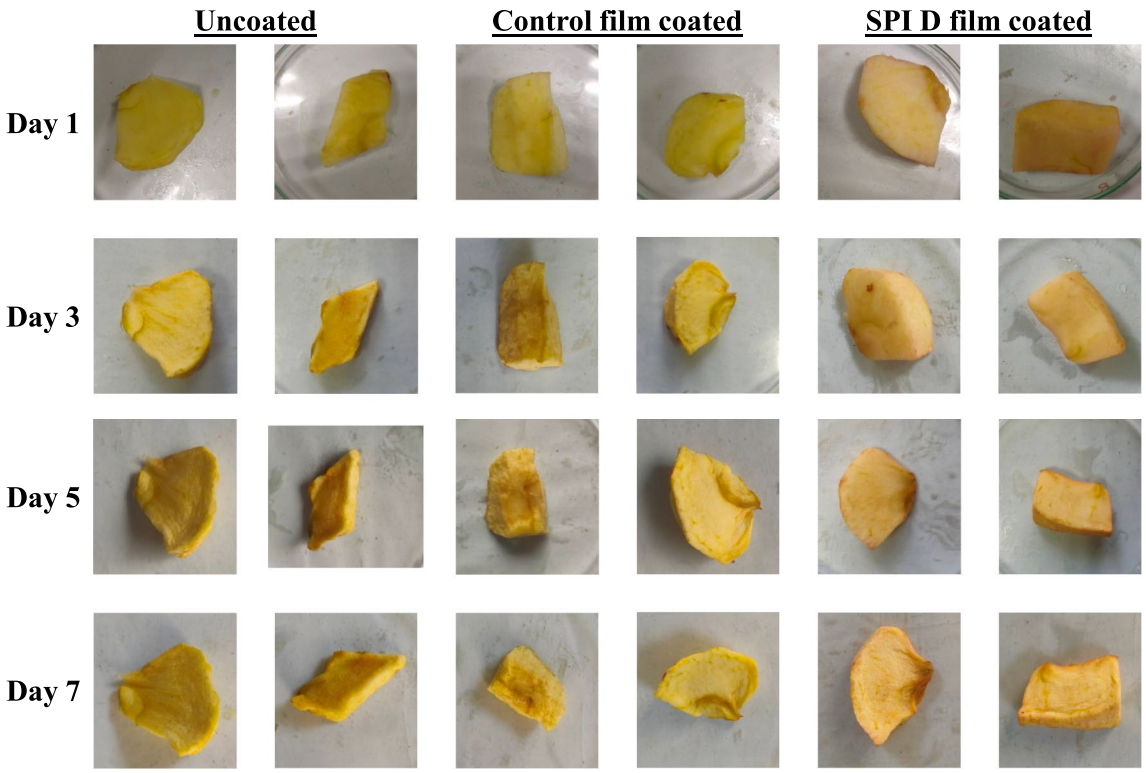


Fig. 4 Effect of different coating on the physical appearance of the freshly cut apples

to be increasing with temperature. For all SPI films, the TPC is higher than the SM limit of 60 mg per kg of simulant. Similar values is found in the literature as well. Haskap leaf extract were incorporated into pea protein isolate films, and TPC release for different films was found to be in the range of 0.67–3.7 mg GAE per/g of film (Chong et al. 2021). Incorporation of blueberry, red grapes, and parsley extract into the chitosan films results in the migration of TPC in the range of 0.12–0.96 mg GAE/g of film Simulant A (aqueous simulant) (Dordevic et al. 2021).

From the migration test, SPI films were found to be more stable in isooctane than 5% ethanol. For 5% ethanol simulant, the OM limit substantially crosses of 60 mg/kg of simulant. Therefore, SPI films have limited application in high water content foodstuffs. However, all SPI films can be used without any restrictions in fatty foodstuffs.

Preservation properties of film coating on freshly cut apples

Weight loss

Minimizing weight loss during storage is one of the main objective of the coating. The effect of extract-incorporated SPI film on weight loss of freshly cut apples was reported for 1, 3, 5, and 7 days as shown in Fig. 3c. The result was compared with uncoated apples as well as SPI control film coated apples. The SPI D film coated apples showed the least weight loss ($P < 0.05$) than uncoated and control film coated apples. This result corresponds with WVP data (Table 1) obtained earlier. Since SPI D film had the lowest WVP compared to other films, so it could have helped reduce the water loss. The antioxidant activity could have also played a role in preventing weight loss. SPI D film showed the highest antioxidant activity, as discussed earlier (Fig. 1a). Alves et al. (2017) studied the preservation effect of ferulic acid incorporated SPI film on freshly cut apples for 7 days. The author reported a similar trend in weight loss of freshly cut apples dip coated with film forming solution. Liu et al. (2017) also observed a similar result of weight loss in freshly cut apples coated with SiOx nano composite SPI film. Furthermore, the improved film property including reduction in weight loss of apples slices as well as the enhanced UV blocking property can be attributed to the blocking by the dragon fruit peel extract particles present in SPI films.

Visual observations

The effect of coating on the physical appearance of freshly cut apples is shown in Fig. 4. The uncoated group showed wrinkles on the surface on day 3 of the storage. While the control film and SPI D film coated group showed wrinkles on day 5 and day 7 of the storage. This could be explained by lower water loss by SPI D film group compared to other groups. Previous study has shown a linear relationship between weight loss and the visual appearance of the fruit (Meindrawan et al. 2018). Moreover, Zhang et al. (2021) reported similar results on the effect of tannic acid loaded chitosan and gelatin coating on the physical appearance of freshly cut apples.

Soluble solids content (SSC)

One essential indicator to assess fruit quality is SSC. All the substances that are soluble in water, such as sugar, acid, vitamins, and minerals, are included in the fruits' soluble solids content (Liu et al. 2017). SSC of the different groups is shown in Table S6. Generally, SSC of all groups decreased with time. Edible coating results in fruit's metabolic activity to decrease. This may cause a slowing down in the polysaccharide degradation process (Liu et al. 2016). The decreasing rate of SSC for SPI D film coated apples was significantly lower ($P < 0.05$) than that of uncoated as well as control film coated apples. This could be explained by the interaction between polyphenol-rich extract in SPI and cell membrane of the apple. This interaction may have an effect on slowing down metabolic and respiratory activity. Similar observations were reported in the literature as well (Liu et al. 2017; Jiang et al. 2021).

PPO activity

As shown in the Table S7, the PPO activity was found to be increasing for all groups of apples from day 1 to day 7. For the uncoated group, the PPO activity increased from 4 ± 1.41 to 11.75 ± 1.71 U min^{-1} ml^{-1}, while for the control film coated group, the PPO activity increased from 3.5 ± 0.71 to 5.75 ± 0.95 U min^{-1} ml^{-1}. For the uncoated group apples, the PPO activity increased to 11.75 ± 1.71 U min^{-1} ml^{-1} on the 7th day of the preservation period. This significant increase in PPO activity could be related to loss in substrate and the creation of quinones on the apple surface due to the oxidation reaction of phenolic compounds. The browning related to PPO activity mainly occurs because of contact of enzymes with phenolic substrates (Zambrano-Zaragoza et al. 2014). Previous study have reported the oxygen barrier property of SPI to be 0.18×10^{11} cm^3 mm m^{-2} day^{-1} atm^{-1} (Liang and Wang 2018). Because of the oxygen barrier property of SPI film, it enables the SPI to act as a barrier to oxygen, which is needed to the browning reaction to take place. Therefore, SPI film coating was found to be better at inhibiting PPO activity than the uncoated apple group. It can be observed that SPI D film coated group had the least increase in PPO activity from 1.5 ± 0.71 to 4 ± 0.81 U min^{-1} ml^{-1}. This effect could be explained by the

presence of total phenolic compounds, betacyanin compounds, and betaxanthin compounds in the RDF extract. These compounds are known to exhibit antioxidant activity as well antimicrobial activity. Similar results were found in other literature as well. Chitosan-based nanoparticle coating over freshly cut apples also inhibited the PPO activity as compared to the uncoated group at 5 ℃ preservation temperature. The authors also reported a similar trend of increase in PPO activity (Pilon et al. 2015). In another study, montmorillonite and citric acid incorporated whey protein isolate film coating over freshly cut apples inhibited the PPO activity (Azevedo et al. 2018).

Conclusion

Findings from this study show that the property of SPI films can be improved by the addition of RDF extract. RDF extract greatly influenced the physical and functional properties of the SPI-based biodegradable film. SPI D film had the highest UV-blocking and antioxidant properties. Cross-linking between extract and SPI was evident from FTIR and SEM results. XRD analysis showed a decrease in film crystallinity, while TGA analysis showed no significant change in the thermal property of the films. Extract addition decreased the WVP and WS values of the biofilm. Furthermore, significant improvement ($P < 0.05$) in tensile strength was reported. However, elasticity was decreased. SPI D film significantly reduced ($P < 0.05$) the weight loss of freshly cut apples compared to uncoated and control film coated apples. Also, the SSC value for SPI D film coated apples was lower ($P < 0.05$) than other groups showing the best preservation effect during 7 days of storage. Therefore, RDF-extract incorporated SPI film can be used as an active packaging material to preserve food products.

Acknowledgements The authors acknowledge the Material Research Center (MRC), MNIT, Jaipur (India), for providing characterization facilities.

Author contributions RAP: Conceptualization, methodology, software, validation, formal analysis, investigation, writing—review & editing; GCJ: Corresponding author, supervision, conceptualization, resources, writing, review and editing, project administration.

Funding This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Data availability Data will be made available on a reasonable request.

Code availability Not applicable.

Declarations

Conflict of interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Ethical approval Not applicable.

Consent to participate Not applicable.

Consent for publication Not applicable.

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